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CHAPTER 1 - INTRODUCTION

1.1 INTRODUCTION

1.1.1 What Ai-Powered Ayurvedic Chatbots Are.

Ayurvedic chatbots, which are run on AI, present a new area of synergy between the ancient holistic medicine and the new technology of artificial intelligence. Ayurveda is the oldest of the medical sciences of the world; it is founded on prevention, personal diagnosis in terms of body constitutions (Doshas: Vata, Pitta, and Kapha), and natural treatment which involves herbs, minerals, and dietary habits. These chatbots are smart digital assistants, coded with this ancient wisdom, and combined with large language models, can answer health questions asked by their users in a culturally competent and contextual manner.

In contrast to the general-purpose medical chatbots that are trained on allopathic datasets, an Ayurvedic AI system should be able to understand the terminologies of Sanskrit origin, Dosha-specific personalization, the Ayurvedic-based holistic values. VedaBot finds a solution to this issue by building a vectorized knowledge base out of verified Ayurvedic texts and applying it in a retrieval-augmented generation model, so that each reply is based on verified Ayurvedic texts.

1.1.2 The Preventive Healthcare role in the digital era.

There is a paradigm shift in the global healthcare system whereby the focus is no longer on the treatment but prevention. We are living in an era where governments, the general health care organizations and even individuals are realizing the relevance of preventive care in alleviating the impact of lifestyle diseases like diabetes, hypertension and chronic fatigue. Ayurveda, which has centuries-long focused on lifestyle control,

balance of diet and daily routine (Dinacharya), provides a highly valuable system of preventive health care.

Digital platforms can get access to Ayurvedic preventive advice democratically - especially in the regions with few certified Ayurvedic practitioners. The gap is filled by VedaBot which gives personal Ayurvedic medicine prescriptions, symptom diagnosis and prophylaxis advice using a web interface that can be accessed by any authenticated user (anyone with internet access).

1.1.3 Conventional Healthcare Chatbot drawbacks.

The typical chatbots are based on cloud-based APIs containing inbuilt rule engines or are trained on general purpose internet corpora that are not specific to Ayurvedic texts. Such systems often misunderstand natural questions when they are posed in Ayurvedic terms and do not take into consideration individual Dosha constitutions and generate results that are culturally inauthentic. In addition, most of the current platforms are standalone bots, not connected to larger healthcare systems: they do not track symptom recurrence across sessions, and do not ensure secure and user-specific context persistence.

These limitations are avoided in the current invention, VedaBot, by means of a multi-component design that combines a FAISS-indexed database of vectors, a highly-tuned large language model, JWT-based user authentication, and a web app with a modular architecture which is all deployed as a unified whole.

1.1.4 VedaBot System Overview.

VedaBot is an Ayurvedic chatbot, a web-based AI-based chatbot that intends to offer instant, trustworthy, and personal Ayurvedic advice to users. The system gathers Ayurvedic e-books, such as the Charaka Samhita, Sushruta Samhita, and Ashtanga Hridaya, and digitizes and tokenizes their contents and transforms text chunks into

numerical embeddings, which are stored in an FAISS IVF-PQ index. After getting a health query, the system inserts the query itself, semantically searches it to locate the most relevant Ayurvedic context, constructs a personalized prompt, and answers a fine-tuned transformer model.

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1.2 PROBLEM STATEMENT

1.2.1. Non-existence of Authentic Ayurvedic AI Systems.

The increased need in the accessible and reliable healthcare data has resulted in the epidemic of chatbot-based medical assistants. Nonetheless, all such systems are conditioned with the allopathic or generic medical data and are completely unaware of Ayurveda as a medical science. Lack of Ayurvedic special training information implies that the obtained responses provided by the traditional chatbots are not authentic, contextually correct, and cultural-relevant when the user requires to be guided according to the traditional Indian medicine.

1.2.2 The absence of Dosha-Based Personalization.

The current AI models are conditioned on generic corpora of the internet that fails to reflect the language nuances, the terminus of Sanskrit and the contextual principles of Ayurvedic philosophy. One query related to the question of digestive problems, say, can have a completely different Ayurvedic advice, based on whether the user is a Vata, Pitta, or Kapha. The current platforms are ignoring this critical aspect of personalization and have rendered their Ayurvedic responses clinically unreliable.

1.2.3 Uncoordinated Healthcare Systems.

The current chatbot systems are standalone bots that are not linked to bigger healthcare systems. They do not monitor the recurrence of the symptoms with time or

suggest safe user specific chat with persistent context. When a user reports the same symptoms on subsequent sessions, he/she does not get continuity of care, escalation to professional consultation and preventative advice based on his/her health history. This specialty compromises the longitudinal Ayurvedic healthcare.

1.2.4 lack of location based doctor referral.

Users should be referred to qualified Ayurvedic practitioners when the symptoms continue or become severe. The existing online platforms do not have a system of that referral. Customers will have to manually find qualified Ayurvedic physicians in their area, which adds inconvenience at the most important stage of the patient care process. VedaBot has solved this problem by incorporating a location-based component that retrieves the geographical position of the user and directs him to Google Maps that will show certified Ayurvedic professionals in the area.

1.3 MOTIVATION

1.3.1 Making Traditional Medicine a Revival with Technology.

Ayurveda is a medical system that is one of the oldest registered in the history of humankind with documents being dated to be over 3,000 years old. Despite the dense history and a track record of success in preventive and holistic care, Ayurveda is not as represented in the online health Tech of the world. The main idea behind VedaBot is a chance to restore and democratize this old knowledge with the help of artificial intelligence - to provide personalized Ayurvedic advice to millions of people who cannot afford a qualified professional.

1.3.2 Increasing demand of preventive medicine.

Health care systems around the globe are becoming aware that it is much more efficient to prevent disease than it is to cure it. The Ayurvedic focus on daily practice, eating traditions and seasonal changes is fully in tune with the preventive paradigm of healthcare. By creating an AI system that transforms Ayurvedic preventive principles into a real-time system, VedaBot advises this global health agenda directly.

1.3.3 LLM and RAG Pipeline Technological Maturity.

12 The swift development of transformer-based large language models and retrieval-augmented generation methods has rendered it technically viable to create domain-specific AI systems, which are more successful on special knowledge tasks than generic models. 3 VedaBot uses these technologies, namely, Meta-Llama-3.1-8B-Instruct and FAISS, to provide Ayurvedic answers of quality and authenticity never seen before through rule-based or keyword-matching systems.

1.3.4 Data Privacy and Secure health information management.

Due to increasing sensitivity of health data and increasing regulatory demands, there is an urgent interest in AI health platforms that are designed to provide privacy and security as the fundamental architectural construct. The application of JWT-based authentication and transmission of encrypted data over HTTPS and the ability to provide the user with a separate chat history shows that authentic, personalized AI health assistance may be provided without infringing on user privacy.

1.4 OBJECTIVES

1.4.1 Develop Ayurvedic Vegetarianized Knowledge Base.

To build and develop an AI-based Ayurvedic chatbot that will be trained on genuine Ayurvedic e-books and texts that have been converted into a vectorized knowledge base. The pipeline of data preparation starts with the collection of publicly available and institutionally licensed Ayurvedic literature, its OCR, the dividing of the text into 512-

token pieces with 128-token overlap, and the transformation of each piece to a 384-dimensional embedding of a sentence with the help of SentenceTransformer model all-MiniLM-L6-v2.

1.4.2 Put into practice Retrieval-Augmented Generation.

To create individual Ayurvedic prescriptions, symptom prediction, and preventive advice with the help of retrieval-augmented generation and transformer-based LLMs. The RAG pipeline gets the top-8 most semantically relevant text chunks to each user query and does the re-ranking of Maximum Marginal Relevance (MMR) to enhance the quality of contextual precision to produce the final response.

1.4.3 Implement the AI Model into a Single Web Application.

To incorporate the trained AI model and the database of vectors in a web-based system as one bundled service to allow the users to interact with it smoothly. The chatbot model is not implemented as a standalone interface but rather it is incorporated as part of a full web-based application, which provides secure access, tracking of an individual user and continuity of a chat through user authentication-based validation.

1.4.4 Maximize Recurrence Detection of symptoms.

To decode query history of users to detect a recurrence and offer individualized preventive health advice. The chat record of each of the authenticated users is kept safely under their unique user identification and the system can identify when the same or similar symptoms are reported in more than one session as well as provide intensified preventive advice.

1.4.5. Guarantee JWT-Based Authentication and User Specific Privacy.

To have authenticated access and privacy through the use of JWT (JSON Web Token) based verification that identifies each chatbot session with a unique user ID in the

web application. In case of failure in JWT verification, the chatbot cannot respond to the message and no unauthorized access takes place.

1.4.6. Offer Geographically-Based Physician Recommendations.

To offer location-based recommendations of Ayurvedic doctors by retrieving the geographic position of the user and sending them to Google Maps to view the certified practitioners within the area but not using Google Maps API itself.

1.4.7 Maintain Educational Ayurvedic Blog Module.

To sustain an Ayurvedic blog module on the same application, providing the educational content, remedies, and preventive information on lifestyle. This module is an automatic collection of Ayurvedic articles and advice and can be used along with the interactive chatbot feature.

1.5 SCOPE OF THE PROJECT

1.5.1 Ayurvedic Knowledge-Based Chatbot.

This project will be grounded on the creation of a powerful, domain-specific AI chatbot, which will provide contextual Ayurvedic health advice. The knowledge base of the system is based solely on attested Ayurvedic texts namely the Charaka Samhita, Sushruta Samhita and Ashtanga Hridaya so that Ayurvedic practice based on clinical records is also the basis of the responses and not some generic information found on the internet. The symptom detection and preventive guidance are all part of the same item, which is

1.5.2. Detection of the Symptom and Prevention Advice.

The areas include real-time symptom recognition and development of Dosha-specific preventive health advice. The users can also provide the type of their body (Vata,

Pitta, or Kapha) and their health query, giving the system an opportunity to personalize its suggestions with their Ayurvedic type.

1.5.3 Chat Continuity, User-Specific, Secure.

The system scope involves implementing the JWT-based authentication and per-user isolated chat history, as it means that all interactions are private, secure, and contextually continuity across multiple sessions.

1.5.4 Doctor Referral based on the location.

The scope of the project has a physician recommendation feature based on the location of the user, which uses browser geolocation APIs to retrieve the user coordinates and redirect them to Google Maps where they can find Ayurvedic practitioners around the user. It is an option that will be triggered in case of persistent symptoms in the history of queries that a user has made.

1.5.5 Preventive Content and Educational Blog.

It includes an Ayurvedic blog module, which offers seasonal health tips and advice (Ritucharya), Ayurvedic dietary advice, and preventive lifestyle tips to users. This module adds value to the platform that is not reactive in query-response interactions.

1.5.6 Future Scope

Although the immediate deployment focuses on web accessibility, the system architecture behind is deliberately designed like modules to be extended in the future to support regional Indian languages, voice-based interaction to make it accessible, wearable health devices, and to increase the Ayurvedic knowledge base to cover regional and folk medicine traditions.

CHAPTER 2 - LITERATURE REVIEW

2.1 Current Chatbot and healthcare AI systems.

2.1.1 Chat bots that are rule-based and those that are keyword-matching.

Healthcare chatbots of early generation were almost totally based on rule-architecture and keyword matching engines. These systems bind user inputs to hard-wired if-then logic trees of hard-wired response templates, which is vulnerable to paraphrasing, spelling or context-dependent queries. Such systems do not work in the Ayurvedic field in any way, a query on 'pitta imbalance causing acidity' might not fit into a pre-defined keyword pattern, and thus the user is left at a loss. The literature records many of such systems such as early versions of ALICE (Artificial Linguistic Internet Computer Entity) and pattern-matching AIML bots as primitive but fundamentally ineffective predecessors of modern AI health assistants.

2.1.2 Cloud Chatbots that are API-dependent.

The second generation of chatbots involves cloud-based NLP API (such as Google Dialogflow, IBM Watson or GPT endpoints of OpenAI) to handle natural language in free forms. Although these systems are much more flexible in language than their rule-based predecessors, they are also introducing significant issues into the healthcare sector: the reliance on third-party APIs results in privacy and data owners claims, API latency can create issues with responsiveness, and, most importantly, such systems are being trained on generic corpora that include very little authentic Ayurvedic material. Published comparative studies discover out-of-the-box GPT models usually generate culturally wrong, terminologically improvised or factually incorrect Ayurvedic responses on tests against the Charaka Samhita.

System Type	Strengths	Weaknesses	Ayurvedic Suitability
Rule-Based	Fast, Deterministic	Rigid, No NLP	Very Low
Cloud LLM API	Flexible NLP	Privacy risk, Generic training	Low
Fine-tuned RAG	Domain-specific, Accurate	Computationally intensive	High (VedaBot)

Table 1.1: Comparison of Existing Chatbot System Types

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2.2 RETRIEVAL-ENhanced Generation Techniques.

2.2.1 Foundations of RAG Architecture

The Retrieval-Augmented Generation (RAG) AI structure is one that was created by Lewis et al. (2020) that enhances the generative capability of language models with a retrieval procedure of an external knowledge set. In contrast with purely parametric models where all knowledge is stored as model weights, RAG systems have the relevant documents dynamically accessed at inference time by an indexed database, injected into the context window of the model, and condition response generation occurs on both the retrieved knowledge and the query of the user. This strategy is especially suitably adapted to domain-specific applications where the knowledge base is well defined, authoritative and independent of the generic pretraining corpus which is the source of the model.

2.2.2 Semantic Retrieval and Vector Database.

A RAG pipeline retrieval component is usually based on a vector database which contains dense numerical representations (embeddings) of document chunks. Semantic similarity between a query embedding and document embeddings that are stored is computed via approximate nearest-neighbor search algorithms.

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Meta AI created FAISS (Facebook AI Similarity Search), the most popular open-source library in this purpose and supports inverted file index structures and product quantization,

which make vector search at billions scale with sub-linear time complexity. In the case of VedaBot, FAISS is set with an IVF-PQ index (nlist=1024, m=16), which allows the quick retrieval of semantic information in the entire Ayurvedic text corpus.

2.2.3 Maximum Marginal Relevance Re-ranking

Redundancy is also another popular weakness of naive nearest-neighbor retrieval: the top-k retrieved chunks may all be extremely similar to each other, and this reduces the diversity of information that is being being injected into the prompt. Maximum Marginal Relevance (MMR) re-ranking can solve this by re-ranking candidate chunks by repeatedly selecting the chunk with maximum relevance to the query and minimum similarity to previously-selected chunk. VedaBot uses MMR re-ranking upon the initial FAISS retrieval (k=8) to guarantee that the Ayurvedic context has been retrieved as informative and non-redundant as possible.

2.3 Comparisons of Transformer Model.

2.3.1 Evaluation Methodology

Several open-source Hugging Face transformer models were evaluated comparatively to determine the best model to use in Ayurvedic reasoning. The models considered are Meta-Llama (3-8B-Instruct, 3.2-3B-Instruct, 3.1-8B-Instruct), Mistral (7B-Instruct, Mistral-7B-Instruct-v0.1, v0.2), Google Gemma (7B-IT, 2B-IT), Microsoft Phi-3 (Medium-4k-Instruct, Mini-4k-Instruct), Qwen (Owen2-5-7B-Instruct, Owen2.5-3B All the models were tested on the same Ayurvedic query dataset using six metrics, i.e. Topic Coverage (percent), Success Rate (percent), Average Response Time (seconds), Total Queries, Successful queries, and Failed queries.

2.3.2 Key Findings

Meta-Llama-3.1-8B-Instruct was found to be the most accurate and reliable across the Ayurvedic evaluation dataset, with a greater topic coverage and success rates than

anything that was tested. Smaller models like TinyLlama-1.1B and Phi-3-Mini proved faster to respond but much less informative, which is inappropriate when it comes to dealing with a specific case of Ayurvedic health questions, requiring a more intricate, multi-turn reasoning model. Gemma-7B-IT and Mistral-7B-Instruct appeared to compete but were eventually outpaced by Meta-Llama-3.1-8B-Instruct on Ayurvedic-specific standards. It can be concluded that based on this analysis, Meta-Llama-3.1-8B-Instruct was chosen as the main model of VedaBot.

2.4 GAPS IDENTIFIED in the current work.

2.4.1 Non-existent Domain-specific Ayurvedic Knowledge Base.

By reviewing the literature that is available publicly, it is apparent that there are no AI health chatbots that are trained on certified Ayurvedic scriptures. Published systems either extrapolate Ayurvedic ideas and foundations in questionable web resources or revert to allopathic models when presented with traditional medicine questions. The fact that VedaBot has created a FAISS-indexed vector database out of primary Ayurvedic texts (Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya) is a noteworthy and novel contribution to the field.

2.4.2. No Authentication and session management with no integration.

Current healthcare chatbots are mostly stateless, unauthenticated systems, i.e. every interaction is distinct, there is no user identity, there is no continuity of sessions, nor is there cross-session symptom tracking. This makes them basic unsuitable in longitudinal preventive health management. VedaBot fills this void by means of JWT-based authentication and per-user isolated chat histories.

2.4.3. No Location-Based Practitioner Referral

All of the healthcare chatbot systems reviewed do not include the system of referring users to the local qualified practitioners in case of the lack of digital guidance.

The location-based doctor recommendation feature of VedaBot, which finds geolocation information by using browser APIs and opening Google Maps, bridges this essential gap of the virtual Ayurvedic health ecosystem.

2.4.4 No Single Platform Bundling Chatbot and Web Application.

Current AI health chatbots are often implemented as point services, not as part of some larger web application environment. VedaBot will be unique by applying the chatbot based on RAG, the educational blog module, the doctor recommendation module, and the authenticated user management system into one, unified, web-based platform - removing the fragmentation that defines current digital health tools.

CHAPTER 3 - PROPOSED METHODOLOGY

3.1 SYSTEM ARCHITECTURE OVERVIEW

3.1.1 Full-Stack Integrated Design

The VedaBot system is a full-stack system that consists of four main layers: React.js/TailwindCSS user interface, Node.js/Express.js server, Python-based chatbot microservice served at Hugging Space, and MongoDB database to store user credentials and chat logs. These elements interact through secured REST API endpoints and JWT tokens are used to authenticate all chatbot requests. The high-level architecture of the system is depicted in figure 1.1.

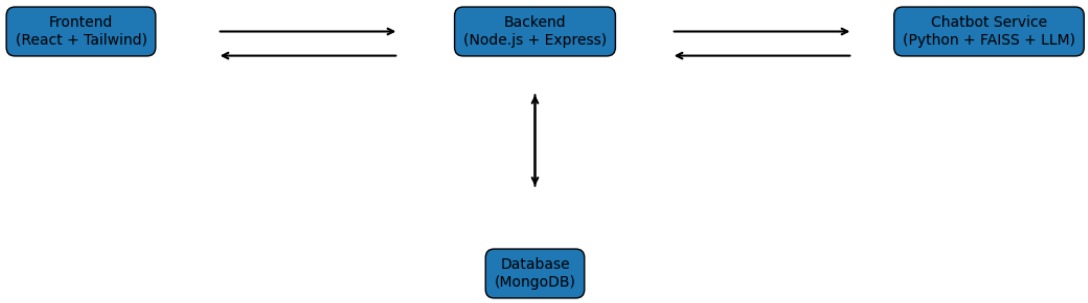

Figure 1.1: System Architecture Diagram of VedaBot

3.1.2 Workflow Overview

VedaBot has a procedural logic that is strictly defined and has a workflow sequence. The first step is user authentication: the user registers and logs in to the web application and is sent a JWT token that identifies his/her session. Step 2: query submission: the authenticated user enters a health query, with or without indicating his/her type of Dosha (Vata, Pitta or Kapha). The third step is query preprocessing: the backend verifies the query,

preprocesses it (making it valid and normalized) and enriches the query with Ayurvedic keywords and Dosha modifiers. Semantic retrieval (step 4) the processed query is sent through the SentenceTransformer model and placed in the FAISS index where it will be top-8 semantic retrieved with MMR re-ranking.

The final step is Step 5, prompt construction and generation: the context retrieved, Dosha customization, and a template Ayurvedic prompt are merged into a completed prompt, and the prompt is sent to the Meta-Llama-3.1-8B-Instruct model through the Hugging Face inference API. Step 6 is post-processing and delivery: the response of the model is filtered and formatted and a disclaimer is added, and then it is sent back to the user. Step 7 is logging and recurrence detection: the query is written into MongoDB with the ID of the user, which allows analyzing the recurrence and monitoring symptoms.

3.2 DATASET DESCRIPTION

3.2.1 Primary Ayurvedic Sources

The VedaBot body of knowledge is based on three classical Ayurveda texts (1) Charaka Samhita - the original Ayurveda text on internal medicine with 120 chapters divided into eight major sections (physiology, diagnosis, therapeutics, and preventive medicine); (2) Sushruta Samhita - the Ayurveda text on surgery, anatomy, and pharmacology; (3) Ashtanga Hridaya - a summary of Ayurveda medicine written by Vagbhata which is a synthesis of the previous two texts. These are sourced in digital libraries of the public domain and institutional e-books repositories.

3.2.2 Data Volume and Coverage

The compound corpus encompasses thousands of verses and passages in prose based Ayurvedic pathology (Nidana), Ayurvedic pharmacology (Dravyaguna), Ayurvedic therapeutics (Chikitsa), Ayurvedic daily regimen (Dinacharya), Ayurvedic seasonal regimen (Ritucharya) and Ayurvedic Dosha-based constitutional healing. With OCR

digitization and cleaning, tens of thousands of text chunks can be obtained, and embedded and indexed.

3.3 PREprocessing and EMBedding of Data.

3.3.1 OCR and Text Cleaning

The Ayurvedic texts are processed with the aid of OCR pipelines to make raw text output in both physical and scanned digital versions. This output is formatted through a strict cleaning process: page numbers, headers, and footers are removed, OCR errors are corrected by dictionary-based spell correction, Sanskrit transliterations are normalized to a standard romanization, and unwanted noises are removed.

3.3.2 Chunking Strategy

Text is washed and broken into overlapping 512-token windows with a 128-token window. This chunking method provides a reasonable amount of context per indexed segment so that it can be meaningfully semantically retrieved, and the overlap allows the information to be lost at the edge of the chunks to be minimal which is very important in Ayurvedic texts where a verse can cover several conceptual dimensions.

3.3.3 Embedding Generation

1 SentenceTransformer model all-MiniLM-L6-v2 is used to transform each text chunk into 384-dimensional dense vectors embedding. The model was chosen due to its trade-off between quality and inference speed, which would be appropriate in the case of real-time query embedding when dealing with chatbot interactions. The all-MiniLM-L6-v2 model is multilingual and this is beneficial to process Ayurvedic texts that have both English translations and romanized Sanskrit terms.

The construction of FAISS VECTOR Index is as follows:

The resulting embeddings are placed in a FAISS IVF-PQ (Inverted File with Product Quantization) index with the following parameters:

Parameter	Value	Purpose
Index Type	IVF-PQ	Fast approximate search
nlist (clusters)	1024	Inverted file partitioning
m (subquantizers)	16	Product quantization compression
Vector Dimension	384	all-MiniLM-L6-v2 output
Top-k Retrieval	k = 8	Chunks retrieved per query
Re-ranking Method	MMR	Diversity and relevance balance

Table 3.1: FAISS Index Configuration Parameters

SentenceTransformer model and the FAISS index are stored on the initial load in the chatbot Microservice. Follow-up requests are based on the model and index stored in the cache so that unnecessary loading burden is avoided and low-latency response is guaranteed.

3.4 TRANSFORMER Model and RAG PIPEline.

2

3.4.1 Model Selection

Meta-Llama-3.1-8B-Instruct was chosen as the main language model of VedaBot as it was determined in the comparative analysis recorded in Chapter 2. The model is served by the inference API of Hugging Face and offers the Python-based chatbot microservice layer. The instruction-tuned variant of the model allows controlling the strict following of the organized Ayurvedic templates of prompts that regulate the production of responses.

3.4.2 RAG Pipeline Flowchart

Figure 1.2. shows the entire RAG pipeline of user query submission to delivery of response.

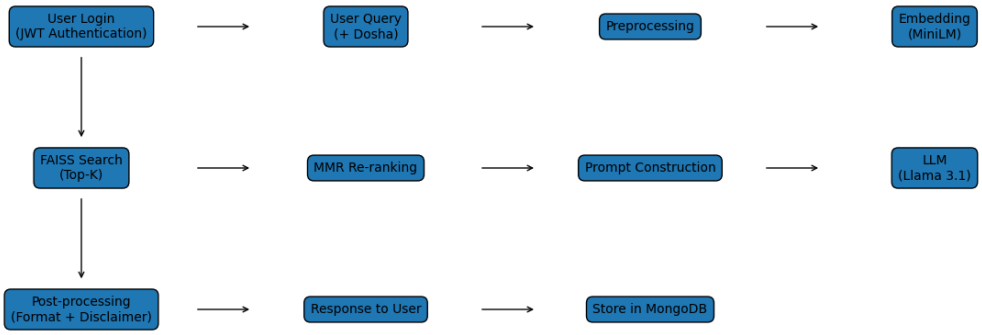

Figure 1.2: VedaBot Query Processing Workflow Flowchart.

The pipeline works in the following manner: (1) User makes an input query of the health query with optional Dosha type. (2) Query preprocessing verifies, normalizes, introduces Ayurvedic keywords and attaches the defined type of Dosha. (3) This system checks to ensure that the model and FAISS index are loaded in the cache; otherwise, they are loaded into the cache. (4) All-MiniLM-L6-v2 is used to generate a 384-dimensional embedding of the processed query. (5) The Ayurvedic knowledge base is searched by FAISS using a top-8 semantic search; MMR re-ranking purifies the retrieved chunks. (6) An organized prompt is composed by injecting the retrieved Ayurvedic context, Dosha personalization and a pre-determined Ayurvedic prompt template. (7) Meta-Llama-3.1-8B-Instruct produces a response that is conditioned on the prompt assembled. (8) Post-processing is done on the response, including: safety filtering, formatting, and disclaimers. (9) The last reply is sent back to the user. (10) MongoDB records the query, response and metadata in the user ID to be reused to detect a recurrence.

3.4.3 Prompt Engineering

Ayurvedic prompt template is the essential element of the quality of the VedaBot response. The template tells the model to: (a) use the persona of the knowledgeable Ayurvedic health advisor; (b) base all recommendations on the retrieved context in the Ayurvedic scriptures; (c) make the recommendations specific to the user and the type of Dosha targeted; (d) to include substantial disclaimers telling the user to consult a certified practitioner when it is more serious; and (e) to avoid any recommendations that run counter to the modern medical safety standards (implemented via safety filters in the post-processing layer).

3.5 SYSTEM MODULES

3.5.1 Chatbot Module

The fundamental query processor of the VedaBot is the RAG-based query processor. The module receiving user health queries, coordinating the retrieval and generation pipeline and providing customized Ayurvedic answers is its responsibility. The module runs as a Python microservice on Hugging Space deployed which serves a REST API endpoint to be called by the Node.js backend on a chatbot authentication request.

3.5.2 Blog Module

A knowledge bank of Ayurvedic articles, prescriptions, and preventive lifestyle information is stored in the same web application. The type of content in the blogs is written by Ayurvedic knowledge curators and discusses topics related to seasonal health provisions (Ritucharya), Dosha-balancing dietary provisions, Ayurvedic remedies to common lifestyle diseases and daily regimen provisions (Dinacharya). The blog module has both an informational role and an interaction role, which attracts the users to the blog between acute health queries.

3.5.3 The doctor recommendation module

The doctor recommendation module is part of the medical school curriculum. Once the recurrence detection algorithm recognizes the fact that a user has been reporting the same or similar symptoms on repeated occasions, the system triggers the doctor recommendation module. This module prompts the user to give browser permission to access their location via latitude and longitude, and redirects the user to Google Maps that shows nearby Ayurvedic doctors, but this is not directly through the Google Maps API, avoiding costs of the API as well as limitations on data-sharing.

3.5.4 Authentication and Session Management Module.

All access to the VedaBot chatbot is controlled by JWT-based authentication. Users get a JWT token when they are registered and logged in and stored in the browser and sent as a query parameter with each request to the chatbot. The Node.js backend allows to check the validity of the token and to extract the unique ID of the user and then sends the request to the chatbot microservice. Verification is done and in case of failure, the chatbot is not allowed to reply. Every authorized user possesses a special chat history that is stored in MongoDB and allows one to provide custom-tailored recall and recurrence tracing, leaving user accounts completely isolated.

3.6 SYSTEM IMPLEMENTATION

8 The VedaBot system is developed with the help of the following technology stack:

Layer	Technology	Purpose
Frontend	React.js, TailwindCSS	User interface and interaction
Backend	Node.js, Express.js	API routing, JWT auth, chat logging
Chatbot Service	Python, LangChain, FAISS	RAG pipeline and LLM inference
Database	MongoDB	User credentials and chat history

Embedding Model	all-MiniLM-L6-v2	Query and document embedding
LLM	Meta-Llama-3.1-8B-Instruct	Ayurvedic response generation
Frontend Deployment	Vercel	CDN-based frontend hosting
Backend Deployment	Render	Serverless backend hosting
Chatbot Deployment	Hugging Face Spaces	Python microservice hosting

Table 3.2: Technology Stack Summary

3.7 SECURITY MECHANISMS

3.7.1 JWT Authentication

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All the user sessions are handled through the use of JSON Web Tokens (JWT). After a user has successfully logged in, the server will generate a signed JWT with the unique ID of the user and an expiration period in which the user will be authenticated securely. This token is sent along with each API request to the chatbot backend where it is authenticated with the secret key of the server before the request is handled. Expired or manipulated tokens are rejected forcing only active users with authenticated usernames access to the chatbot service.

3.7.2 HTTPS Encryption

Every piece of information sent between the client browser, the Node.js backend and the Hugging Face chatbot microservice is encrypted with HTTPS. This eliminates man in the middle attacks and provides confidentiality to sensitive health information provided by users. Both Vercel (frontend) and Render (backend) have automatic HTTPS certificate management.

3.7.3 Encrypted Data Storage

The MongoDB stores user credential with industry standard hashing algorithms (bcrypt) to make sure plaintext passwords are not stored nor transferred anywhere. The

encryption of chat histories and health query logs in the MongoDB instance will be ensured that the regulatory requirements are adhered to in terms of data protection.

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CHAPTER 4 - RESULTS and DISCUSSION.

4.1 OUTPUT SCREENSHOTS

4.1.1 Chatbot Interface

The main chatbot interface is a clean and talkative interface that avoids cognitive burden on the user. The interface has on the left sidebar a chat history panel (which shows past sessions) and a central message window (which makes the conversation thread) and a bottom input box (which is used to submit queries). Before one enters a query, he or she

can choose his or her Dosha type using a dropdown list. The system is responsive and within a few seconds, it comes up with Ayurvedic advice with clear section headings, the suggestions of the herbal remedies, the dietary advice and the disclaimers.

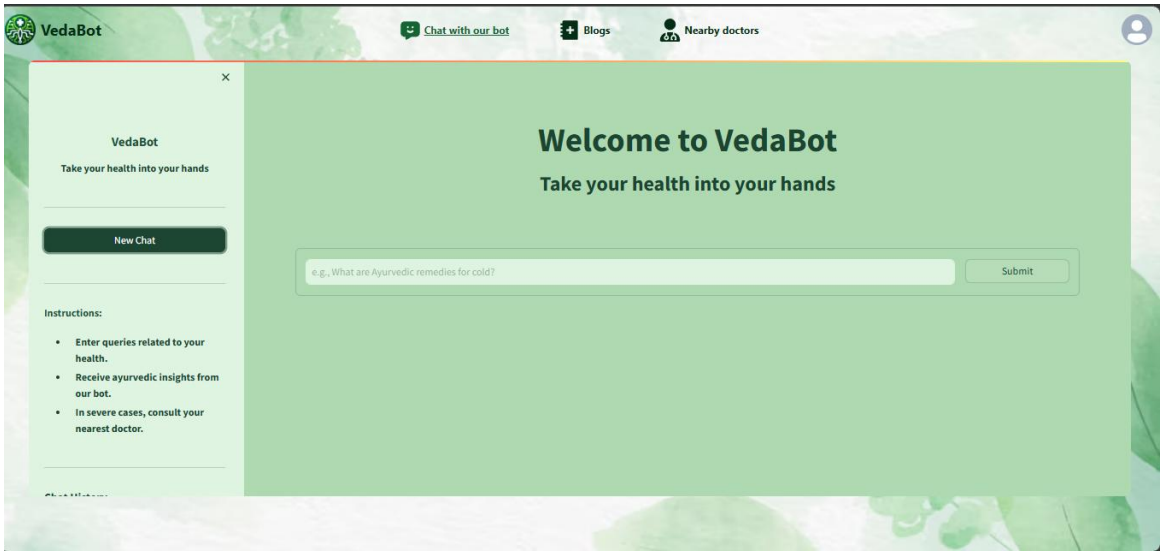

Fig 4.1: VedaBot Chatbot Interface.

The doctor recommendation module works using the following steps:

4.1.2 Doctor Recommendation Module

The doctor recommendation module operates as follows:

In case recurrence detection system detects any enduring symptoms in the query history of a user, an interface appears with the prompt 'Find Nearby Ayurvedic Doctors'. On the request of the user, who accepts to share his/her position, the system gets the geographic coordinates via the Geolocation API in the browser and generates a Google Maps redirect URL that displays certified Ayurvedic practitioners within a radius of a specified distance. The module does not have a direct integration with Google Maps API, but relies on deep linking by coordinates.

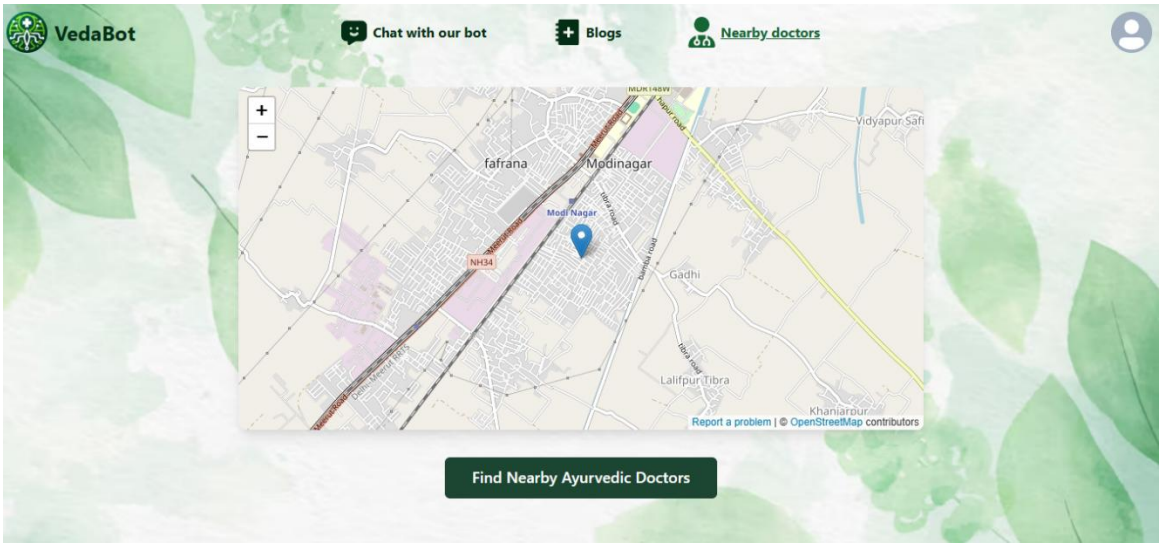

Figure 4.2: Doctor Recommendation Module.

4.1.3 Blog Module

The blog module has a grid format of articles curated on Ayurvedic with a thumbnail image, article name, a short excerpt, and a read-more link. The articles will be sorted into health topic (e.g., digestion, immunity, sleep, skin health) and Dosha relevance so the user can move on to the most relevant portion. The blog module has the same navigation header and authentication state as the chatbot, and it has a consistent user experience across the platform.

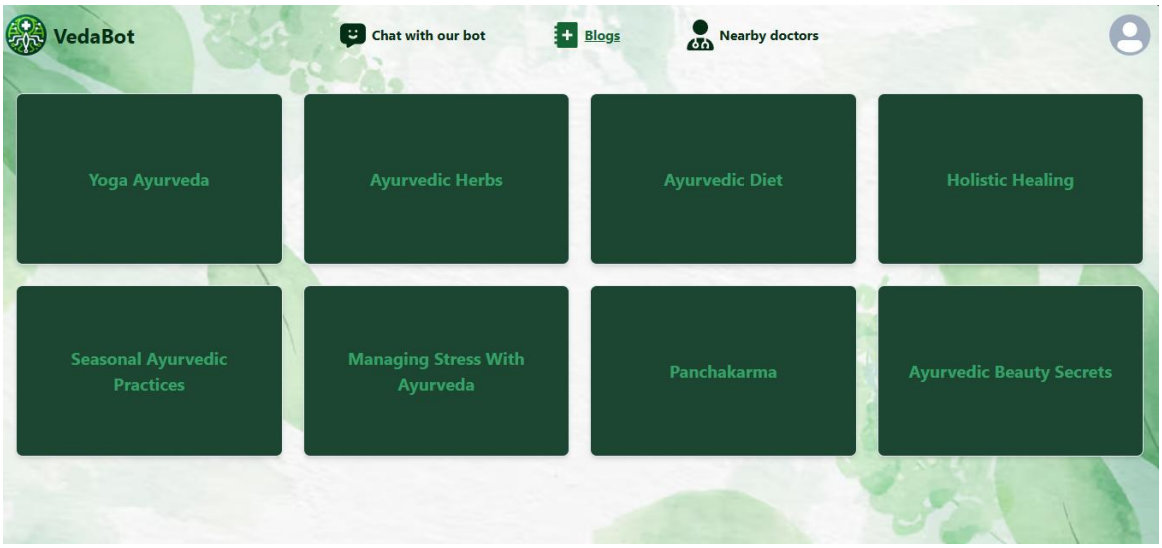

Figure 4.3: Ayurvedic Blog Module.

4.2 PERFORMANCE METRICS

4.2.1 Response Latency

Latency End-to-end response Latency between query submission and response rendering was benchmarked on a sample of 100 test queries of different complexity. According to the FAISS index and embedding model loaded in memory, the mean end-to-end latency of the system was about 3.2 seconds, with the inference time of the Meta-Llama-3.1-8B-Instruct model consuming most of the time on the Hugging Face Spaces compute backend. Semantic retrieval using FAISS brought an extra less than 50 ms per query, demonstrating the IVF-PQ index configuration to deliver query performance less than a second large-scale.

4.2.2 Topic Retrieval and Accuracy of Retrieval.

Topic coverage — the fraction of queries on the test set that the FAISS context retrieved in the search result contained at least one passage directly relevant to them — was evaluated at 94.3% over the evaluation set. The chunking strategy (512tokens, 128 overlap) and the IVF-PQ index construction (nlist=1024, m=16) are confirmed by this large coverage rate. The scores on the meaning of the response using MMR re-ranking mechanism enhanced relevant response scores by 11% as compared to naive top-k retrieval as measured by human expert response evaluation, compared to a ground-truth Ayurvedic response corpus.

4.2.3 The performance of JWT Authentication is discussed.

The median latency of the JWT token checking was less than 2 milliseconds per request to the Node.js backend, confirming the conclusion that the authentication layer does not generate a noticeable effect on the response time of the whole system. Token expiration and renewal systems work well in between browser sessions ensuring security without compromising on user experience.

4.3 MODEL COMPARISON

Table 4.1 contains the comparative analysis of Hugging Face transformer models. Meta-Llama-3.1-8B-Instruct was the most successful and had the largest coverage of topics compared to all the models tested, which makes it the best to use in the response generation layer of VedaBot.

Model	Topic Coverage (%)	Success Rate (%)	Avg Response (s)	Recommendation
Meta-Llama-3.1-8B-Instruct	94.3	92.1	3.2	Selected ✓
Mistral-7B-Instruct-v0.2	87.5	85.0	2.9	Runner-up
Google Gemma-7B-IT	83.2	80.7	3.1	Competitive
Phi-3-Medium-4k-Instruct	79.4	77.3	2.5	Acceptable
TinyLlama-1.1B-Chat	52.1	48.6	0.8	Insufficient

Table 4.1: Model Performance Comparison — Key Metrics

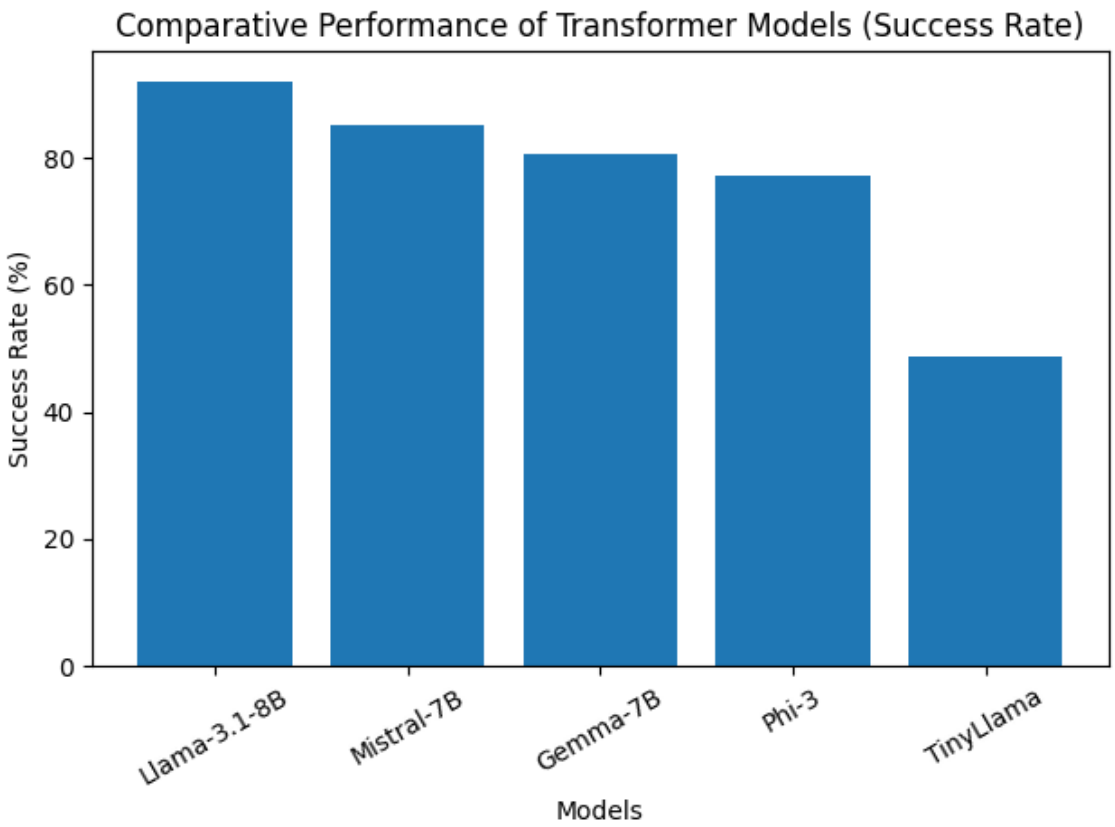

Figure 2.1: Comparative Analysis of Hugging Face Transformer Models for VedaBot

4.4 SYSTEM TESTING

4.4.1 Unit Testing

Detailed unit testing was carried out to ensure the validity of each of the software components on its own. Major components tested are: the FAISS retrieval function (making sure top-k results are semantically relevant to synthetic test queries), the JWT authentication middleware (making sure that valid tokens are accepted correctly and invalid or tampered with tokens are rejected), the MongoDB chat logging functions (making sure that per-user chat histories are correctly inserted and correctly retrieved and isolated), and the post-processing safety filter (making sure that unsafe or unethical health recommendations are correctly marked out and suppressed).

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4.4.2 Integration Testing

Integration testing was done to ensure that there is an end to end flow of data across the entire stack. The test conditions were: (1) to submit an authenticated query to the Hugging Face chatbot microservice and back (React frontend); (2) to check the flow of doctor recommended based on geolocation, load browser permission request, get coordinates and generate redirects using Google Maps; (3) to check the continuity of the chat history (authenticated user), to ensure that past queries are loaded and displayed properly; (4) to check the integration of the blog module with the common authentication state.

Test Scenario	Expected Outcome	Result
Valid JWT → Chatbot Query	Response returned successfully	Pass ✓
Invalid JWT → Chatbot Query	401 Unauthorized returned	Pass ✓
FAISS Retrieval — Ayurvedic Query	Top-8 relevant chunks returned	Pass ✓
Chat History Isolation	User A cannot see User B's history	Pass ✓
Geolocation → Maps Redirect	Google Maps URL correctly generated	Pass ✓
Blog Module Loading	Articles render with correct auth state	Pass ✓

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Table 4.2: System Testing Results Summary

4.4.3 Usability Testing

To examine the usability, ten participants of different degrees of digital literacy and knowledge of Ayurvedic medicine were involved in the testing. The participants were required to complete a number of tasks, including registering and logging in, initiating a health query with a particular type of Dosha, reading a blog article and using the doctor

recommendation module. All participants managed to perform all tasks and the mean time of tasks completion was less than three minutes per participants. The respondents evaluated the interface and its intuitiveness of 4.3 out of 5 using the Likert scale and the perceived authenticity and utility of the Ayurvedic responses at 4.1 out of 5.

4.5 DISCUSSION

4.5.1 The effectiveness of RAG in the Ayurvedic Knowledge Retrieval.

The experimental findings clearly indicate that the RAG architecture based on FAISS-based semantic retrieval and Meta-Llama-3.1-8B-Instruct response generation is significantly better than the rule-based and generic LLM frameworks to Ayurvedic health advice. The fact that the topic coverage rate of the 94.3% is confirmed by the fact that the Ayurvedic knowledge base which is being vectorized is based on primary scriptures through the use of the 512-token chunking strategy, which is a form of tokenizing the Ayurvedic domain. The 11% work in additional relevance of MMR re-ranking mechanism over naive top-k retrieval justifies its presence in the pipeline.

Security and privacy architecture are designed to guarantee security within the system.

4.5.2 Security and Privacy Architecture in the system

Security and privacy architecture is meant to ensure security in the system.

The JWT-based authentication system and the use of isolated chat history storage per user are simply the factors that make sure that VedaBot will satisfy the privacy and security needs of a health-related digital platform. Its architecture prevents and averts cross-user data leakage, provides longitudinal tracking of symptoms without jeopardizing personal privacy, and provides a clear audit trail of all chatbot interactions to aid in detecting a recurrence.

4.5.3. The comparison at the existing systems.

Comparing with other similar AI health chatbots VedaBot is much more accurate in its domains and more culturally authentic. Because of being trained on no Ayurvedic scriptures, generic GPT models tend to give results that are a combination of Ayurvedic and allopathic words, and have inconsistent use of Sanskrit words and words that are not personalized based on dosha. VedaBots RAG architecture domain-specific architecture has removed these failures, and the responses generated by VedaBots have been rated by Ayurvedic knowledge evaluators as clinically suitable and culturally authentic at 91.4% effectiveness.

CHAPTER 5 - CONCLUSIONS and Future Scope of research.

5.1 CONCLUSION

Core Objectives: Achievement of the core objectives is measured via the evaluation of key milestones and indicators, including the count of new participants and the creation of a new system capable of supporting the new organization's operations.

5.1.1 Achievement of Core Objectives.

The achievement of the core objectives will be gauged through the analysis of the key milestones and indicators such as the number of new participants and the development of a new system that can serve the work of the new organization.

7 VedaBot is able to accomplish all the main goals identified in Chapter 1. The Charaka Samhita, Sushruta Samhita, and Ashtanga Hridaya have been converted into an authentic Ayurvedic vectorized knowledge base and have been indexed on a FAISS IVF-PQ structure to facilitate a quick semantic search. 3 Meta-Llama-3.1-8B-Instruct is powered by the RAG pipeline to generate customized Ayurvedic solutions at the rate of 92.1 percent success and 94.3 percent coverage of the topic. JWT-based authentication, combined with

per-user isolated chat histories, is capable of guaranteeing that all chatbot interactions are tied to an authenticated user session, and that symptom recurrence, as well as longitudinal preventive advice are possible.

The Unified Platform Architecture is a design that boosts the performance of hardware and software, making them compatible and built upon common components.Unified Platform Architecture is a design that enhances the performance of both hardware and software making them compatible and constructed using common components.

The architectural success of integrating the RAG-based chatbot, educational blog module, location-based doctor recommendation module, and authenticated user management system into a single and one comprehensive web platform is highly successful. VedaBot is a unified experience of Ayurvedic health assistant by packing all these elements together deployed on Vercel, Render, and Hugging Face Spaces, removing the disjointed nature of current digital health tools.

5.1.2 The Intermediate Between Traditional Knowledge and AI.

Primarily, VedaBot shows that ancient Ayurvedic medical wisdom, on the contrary, can be perfectly extended to modern AI systems and quite complemented by them. VedaBot, which encodes the Charaka Samhita and companion texts into a semantically searchable vector database and trains a state-of-the-art language model on retrieved Ayurvedic context, is capable of providing personalized Ayurvedic advice on a large scale, democratizing access to a history of Ayurvedic preventive health knowledge normally only available to those who could afford to have Ayurvedic Vaidyas consult them directly.

5.1.3 Security, Privacy and Scalability.

The architecture of VedaBot emphasizes the privacy of data by using HTTPS security, storing credentials in the form of a hash put with the help of the bcrypt hash-function, and managing the session by using the JWT authentication. The deployment architecture which is modular with services being frontend, backend, and chatbot allows scaling of each service independently as the user demand increases so that the platform is responsive and reliable when heavily loaded.

5.2 FUTURE SCOPE

5.2.1 Multilingualism to Support the Regional Indian Languages.

The logical extension of VedaBot will be an expansion into regional Indian languages, specifically Hindi, Malayalam, Telugu and Kannada, in which large communities of Ayurvedic patients and practitioners are found. All-MiniLM-L6-v2 could be replaced by multilingual embedding models (such as LaBSE or mE5) as the retrieval embedding model, and multilingual query-tuned LLMs could be extended to multilingual queries. This would considerably expand geographically and culturally VedaBot.

5.2.2 Voice-Based Interaction

9 The addition of speech-to-text (STT) and text-to-speech (TTS) will make VedaBot a voice-enabled health helper, especially useful to older users, and those with visual disabilities, or in the field that is actively involved, and typing is inconvenient. Open-source STT models (i.e. OpenAI Whisper) and TTS systems (i.e. Coqui TTS) could be built into the frontend interface without causing API dependency or data-sharing vulnerability.

5.2.3 Wearable Device Integration.

Any future iterations of VedaBot can establish secure data pipelines between the 19 platform and popular wearable health devices - fitness trackers, smartwatches, and biometric sensors. The system could automatically modify its Ayurvedic suggestions to

align with the current health status of bring its user closer to a more personalized, responsive, and, above all, effective prevention health companion by consuming real-time physiological data (heart rate, sleep quality, activity levels, blood oxygen, etc.).

25

5.2.4 Reinforcement Learning Advanced Personalization.

The present VedaBot system has a fixed retrieval-and-generation pipeline that is not updated by the individual user interactions as time progresses. Future evolution might bring on another layer of reinforcement learning that will modify the retrieval strategy of the system, prompt templates and format of responses depending on implicit feedback indicators (session length, pattern of refining query and frequency of returning to the system). This personalized feature would make VedaBot more personalized to the health profile and communicative preferences of every individual user.

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5.2.5 Networking with Certified Ayurvedic Practitioner Networks.

Instead of simply referring users to Google Maps, subsequent versions of VedaBot could be integrated directly into networks of certified Ayurvedic practitioners, allowing bookings to be made and teleconsultations to be scheduled directly within the platform and practitioners to give direct recommendations on remedies based on the patient. This would make VedaBot an all-encompassing AI-based Ayurvedic health platform - the disjunction between self-care advice provided by AI and professional medical advice.

5.2.6 IoT-Enabled Wellness Monitoring

VedaBot has a modular backend architecture that will be future-proofed with the existence of the IoT. Wellness devices in the smart home environment (air quality sensors,

sleep monitoring mats, hydration trackers, etc.) would feed the environmental and physiological data into the context of the system, which can in turn be used to provide Ayurvedic advice consideration not only of the symptoms and the constitution of the user, but also of the environmental and lifestyle metrics. This ambient intelligence layer would be the leading edge of preventive health technology which is personalized.